Akcelport industries

ISO 9001 & 45001 Certified Company

CABLE DUCTS









WIRING DUCTS

13, Shah Industrial Estate, Deonar, Chembur, Mumbai 400088, India
Tel : +91 22 2555 5888 email : mail@akcelport.com Web : www.akcelport.com

CABLE DUCTS

Features

- Manufactured from high impact rigid polyvinyl chloride.
- Non-flammable, non-corrosive and rust proof
- Will not peel, chip or crack
- High dielectric strength, high temperature withstanding capacity
- Side slots are designed to prevent popping up of wires
- Heavy and robust design of duct and cover for efficient and fast locking

Colour

Standard colour is grey.

Other colours – Black, White, Ivory, Green, Blue are available for bulk quantities

RAW MATERIAL PROPERTY DATA SHEET

PVC FRLS CABLE DUCTS TESTS				
Test	**Test Purpose**	**Test Standard**	**Criteria**	**Typical Values**
Limiting Oxygen Index (LOI)	Minimum Oxygen Concentration for supporting combustion of wiring duct insulation at normal temperature	ASTM D 2863	Min. 29	Min. 40
Temperature Index	Temperature index at which normal oxygen content (21%) of air will support combustion of wiring duct material	ASTM D 2863	>200°C	>200°C
Maximum Stroke Density Rating (SDR)	Indicates the visibility under fire	ASTM D 2863	<60%	<45%
Flammability Class	Should be highly flame retardant	UL-94	V-0	Passed

CABLE DUCTS

PVC FRLS WIRING DUCT : RAW MATERIAL SPECS			
Properties	**Units**	**Test Standard**	**Typical Values**
General Characteristics			
Specific Gravity	g/cc	ASTM D 792	<1.48
Heat Deflection Temperature (HDT) @18.2 Mpa	°F	ASTM D 648	Upto 160
Vicat softening point (VSP) at 5kg load	°F	ASTM D 1525	Upto 185
Water absorption	%	ASTM D 570	0.045
Hardness -Shore	D	D ASTM D 2240	Max. 78
Burning Characteristics			
Flammability Class	-	UL-94	V-0
Oxygen Index Test (LOI)	%	ASTM D 2863	Min. 40
Maximum Smoke Density Rating (SDR)	%	ASTM D 2843	<50
Mechanical Characteristics			
Tensile Modulus	Kg/cm^2	ASTM D 638	>12000
Tensile Strength at break	Kg/cm^2	ASTM D 638	>390
Izod Impact Strength (Notched)	Kg.cm/cm	ASTM D 256	>7.0
Compressive Strength	Kg/cm^2	ASTM D 695	>1100
Electrical Characteristics			
Power Factor	-	ASTM D 150	0.018
Dielectric Constant	-	ASTM D 150	<3
Dielectric Strength	KV/mm	ASTM D 149	>12
Surface Resistivity	Ohm	ASTM D 257	>1x1015
Volume Resistivity	Ohm-cm	ASTM D 257	>3x1016

HALOGEN FREE - FR DUCTS : RAW MATERIAL SPECS			
Properties	**Units**	**Test Standard**	**Typical Values**
General Characteristics			
Specific Gravity	g/cc	ASTM D 792	<1.2
Heat Deflection Temperature (HDT) @4.6 kg/cm^2	°F	ASTM D 648	Upto 220
Vicat softening point (VSP) at 5kg load	°F	ASTM D 1525	Upto 240
Co-efficient of linear thermal expansion	°C	ASTM D 696	7.2×10^{-5}
Water absorption	%	ASTM D 570	0.6
Hardness – Rockwell R	R	ISO 2039-2	124
Burning Characteristics			
Flammability Class	-	UL-94	V-0
Limiting Oxygen Index (LOI)	%	ASTM D 2863	Min. 35
Maximum Smoke Density Rating (SDR)	%	ASTM D 2843	Max. 25
Mechanical Characteristics			
Tensile Modulus	Kg/cm^2	ASTM D 638	>2500
Tensile Strength at break	Kg/cm^2	ASTM D 638	>525
Izod Impact Strength (Notched)	cm-kgf/cm	ASTM D 256	Min. 65
Flexural Modulus	Kg f/cm^2	ASTM D 695	Min. 25000
Electrical Characteristics			
Dielectric Strength	KV/mm	ASTM D 149	>12
Surface Resistivity	Ohm	ASTM D 257	>1x1015
Volume Resistivity	Ohm-cm	ASTM D 257	>3x1016

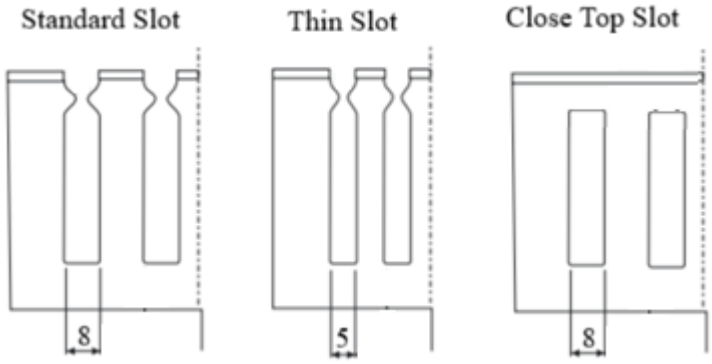
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SLOTTING DIMENSIONS

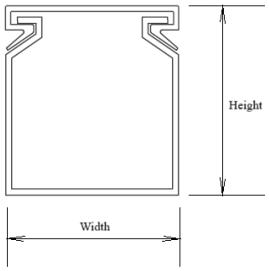


THIN SLOT (T)
Slot Distance : 5mm
Distance Between Slot : 8mm
Includes Wire Retaining Grooves above size 60mm

STANDARD SLOT (S)
Slot Distance : 8mm
Distance Between Slot : 12mm

CLOSE TOP SLOT (U)
Slot Distance : 8mm
Distance Between Slot : 8mm

Wiring Ducts

DUCT PROFILE TYPE	ITEM	Width mm	Height mm	Available Slotting Styles
	25X25	25	25	T, U
	25X30	25	30	T, U
	25X40	25	40	T, S, U
	25X45	25	45	T, S, U
	25X60	25	60	T, S, U
	25X80	25	80	T, S, U
	40X40	40	40	T, S, U
	40X60	40	60	T, S, U
	40X80	40	80	T, S, U
	45X45	45	45	T, S, U
	45X60	45	60	T, S, U
	45X80	45	80	T, S, U
	60X40	60	40	T, S, U
	60X45	60	45	T, S, U
	60X60	60	60	T, S, U
	60X80	60	80	T, S, U
	80X80	80	80	T, S, U
	100X100	100	100	T, S
	150X100	150	100	T, S

Formula To Select Wire Duct For Wire

Approx wire Fill Capacity = 50% of $\left(\frac{\text{Usable Duct Area}}{\text{Wire Area}}\right)$

Usable Duct Area = W x H x 0.90

Wire Area = 0.785 x D^2 or $\Pi \times \frac{D^2}{4}$

W = Width of Duct in mm H = Height of Duct in mm

D = Diameter of Duct in mm

Example : Duct size = 45mm (H) x 45mm (W)
Diameter of wire = 3.10mm.

Wire fill capacity = $0.50 \times \left(\frac{45 \times 45 \times 0.90}{0.785 \times 3.10 \times 3.10}\right)$

Wire fill capacity = $0.50 \times \left(\frac{1822.5}{7.544}\right)$

Wire fill capacity = 0.50 x 241

Approx Wire fill capacity = 120 wires

Special Halogen Free Ducts also available with us

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